

It is here that the assumption of normality helps us. Let us consider it. From Section 9.6, we know that  $\hat{\beta} = \sum kY$  and thus, it is a linear function of the observed values of  $Y$ . But,  $Y = \alpha + \beta X + U$ . Consequently, we can see,  $\hat{\beta} = \sum k(\alpha + \beta X + U)$ . Now, we already know that the  $k$ s, the  $X$ s and the parameters are all given. As a result, finally,  $\hat{\beta}$  becomes a linear function of the random disturbance variable  $U$ . Thus,  $\hat{\beta}$  has the same distribution as  $U$ . Therefore, the assumption of normality  $U$ , implies that  $\hat{\beta}$  is distributed as normal.

We have already seen that  $E(\hat{\beta}) = \beta$  and  $se(\hat{\beta}) = \frac{c}{\sqrt{\sum (X - \bar{X})^2}}$ . It follows

then that

$$\hat{\beta} \sim N \left( \beta, \frac{\sigma}{\sqrt{\sum (X - \bar{X})^2}} \right)$$

You must have learnt from your compulsory course in quantitative methods, that now a standard normal variable can be formed with mean equal to zero and standard deviation equal to one and it can be used for testing of hypotheses regarding  $\beta$ . But the problem here is that the population standard deviation  $c$  is unknown and we have to use an estimate for it. However, we know that an

unbiased estimate of  $c$  is given by  $\hat{\sigma}^2 = \frac{\sum \hat{U}^2}{n-2}$ , where,  $\hat{U}$  is the sample regression error term and is therefore, computable, and  $n$  is the size of the sample. Again you must be knowing that when we standardize  $\hat{\beta}$  by subtracting its mean from it and divide it by its estimated standard deviation, it no longer follows normal distribution and it in fact follows a *student-t* distribution with  $n-2$  degrees of freedom. This *student-t* distribution has a mean equal to zero and a standard deviation equal to one. Thus, it is this standard *student-t* distribution that is used for testing of hypothesis about  $\beta$ . We have a table for such a standard *student-t* distribution for different degrees of freedom. And like the standard normal table, this table is put into use for such tests of hypotheses.

Let us see now how we can conduct some test of hypothesis regarding the unknown population regression coefficient by considering some examples that we have already discussed in Section 9.7. Consider Example 9.2. In this example, we have considered the issue of the consumption function in the Indian economy by regressing final consumption expenditure on GDP. To begin with, we might be interested in examining whether the relationship is significant or not. Thus, our focus is on the regression coefficient  $\beta$  and by the nature of the inquiry, we have to conduct a two-tailed test. We can proceed by setting our null hypothesis and the alternative hypothesis in the following manner:

|                        |                      |
|------------------------|----------------------|
| Null hypothesis        | $H_0 : \beta = 0$    |
| Alternative hypothesis | $H_1 : \beta \neq 0$ |

The acceptance of the null hypothesis implies the rejection of the alternative hypothesis and this in turn implies that on the basis of the sample information there does not seem to be any significant relationship between GDP and the final consumption expenditure. On the other hand, the rejection of the null hypothesis implies the acceptance of the alternative hypothesis and this in turn implies that on the basis of the sample information there seems to exist significant relationship between GDP and the final consumption expenditure. The next step is to construct our test statistic, which is conventionally denoted by  $t$ . Thus

$$t = \frac{|\hat{\beta} - E(\hat{\beta})|}{s.e(\hat{\beta})}$$

Under the null hypothesis,

$$t = \frac{|\hat{\beta} - 0|}{\sqrt{\sum \frac{\hat{U}^2}{n-2}}} = \frac{|\hat{\beta}|}{\sqrt{\sum \frac{\hat{U}^2}{n-2}}}$$

And this is distributed as a *student-t* with  $n-2$  degrees of freedom. From the results of the Example 9.2 we can easily compute the value of this  $t$  statistic since, both the values of the estimated regression coefficient (0.719676) and the standard error of the estimated regression coefficient (0.007865) are mentioned there. In fact, even the computed value of the  $t$  statistic (91.50314) is given. In addition, the degree of freedom is also given (20). Normally, any test of hypothesis is conducted either at 1 per cent level of significance or at 5% level of significance. From the student-t distribution table we can find out the critical values of the test statistic  $t$  for both 1 per cent level of significance and at 5% level of significance at a degree of freedom of 20 for two-tailed test. The values are 2.845 and 2.086 respectively. Thus, the computed value of  $t$  far exceeds both the critical values. Thus, on the basis of the sample information we cannot accept the null hypothesis. This amounts to accepting the alternative hypothesis. Consequently, personal consumption expenditure does seem to be dependent upon GNP in India during the sample period 1980-2001.

Once the relationship between GDP and personal consumption expenditure has been established some further test can be conducted about the likely value of the marginal propensity to consume out of GNP. For example we can conduct a test regarding whether the marginal propensity to consume is 80 per cent or not. Thus we have

$$\text{Null hypothesis} \quad H_0 : \beta = 0.80$$

$$\text{Alternative hypothesis} \quad H_1 : \beta \neq 0.80$$

Thus our  $t$  statistic in this case is given by

$$t = \frac{|\hat{\beta} - E(\hat{\beta})|}{s.e(\hat{\beta})} = \frac{|0.719676 - 0.80|}{0.007865} = \frac{|-0.080324|}{0.007865} = 10.212841.$$

Now this value of the test statistic also exceeds the critical values of 2.845 and 2.086 for a degree of freedom of 20 at 1 per cent and 5 per cent levels of significance respectively. Thus, on the basis of the sample information the

difference between estimated value of  $\beta$  and its mean is so much that even in 1 out of 100 cases or 5 out of 100 cases we do not expect to obtain such a difference. Thus on the basis of the sample information, we are not in a position to accept the null hypothesis. Hence, in all probability during the sample period of 1980-2001, India's marginal propensity to consume out GDP has not been as high as 80 per cent.

In the above example, we considered two-tailed tests of hypotheses. This however, does not rule out the possibility of conducting one-tailed tests. It all depends upon the type of inquiry that we intend conducting.

Thus, a discussion on the test of hypotheses finally marks the end of our discussion on two variable regression model.

### Check Your Progress 2

- 1) When will you use a log linear regression model?

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- 2) How do you interpret the estimated slope coefficient of a log linear regression model?

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- 3) If you want to estimate India's rate of growth of per capita income during the period 1980-2000, what should be the functional form of your regression model?

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- 4) Explain the concept of a classical normal regression model. What is its use?

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- 5) Explain briefly the steps for testing the significance of the regression coefficient.

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### 9.11 LET US SUM UP

Our focus has been exclusively on the two variable regression model. The purpose of this unit has been to specify what is known as the classical regression model in the literature for the population. It is essentially a linear in parameter regression model. This regression model is stochastic in nature in the sense that here the dependent variable is taken as a random variable as against the independent variable being considered as non-random in nature. The element of stochasticity is introduced by including a random error or disturbance term in the model. Our major concern has been to estimate the unknown parameters of the population regression model from the known sample information and tests some hypotheses about these parameters. In this regression model we usually make five assumptions about the disturbance term in order to effectively estimate the parameters by implementing the ordinary least square procedure. The estimates thus obtained satisfy some desirable properties as enunciated in the Gauss-Markov Theorem. In this connection, it may be mentioned here that if some of these assumptions are not fulfilled, some problems regarding the sample estimates of these unknown parameters arise. These problems and their solutions have not been discussed in this unit. We have also considered different functional forms of the regression model that might be relevant to use in different situations. For the purpose of hypothesis testing, we have to extend the assumptions of the classical regression model to include one more assumption about the probability distribution of the disturbance term. The resultant regression model is then known as the classical normal regression model. We have considered the use of this model for the tests of hypotheses.

## 9.12 EXERCISES

- 1) The following table presents the data for broad money supply  $M_3$  (Rs. Crore) for India during 1980-2002. Fit in an appropriate regression model and estimate the rate of growth of money supply in India during this period.

**Table 9.5: India's Broad Money Supply 1980-2002**

| Year | M3      |
|------|---------|
| 1980 | 50966   |
| 1981 | 59793   |
| 1982 | 68515   |
| 1983 | 80577   |
| 1984 | 95295   |
| 1985 | 111096  |
| 1986 | 130653  |
| 1987 | 153207  |
| 1988 | 179687  |
| 1989 | 213856  |
| 1990 | 249493  |
| 1991 | 292403  |
| 1992 | 344238  |
| 1993 | 399048  |
| 1994 | 478196  |
| 1995 | 552953  |
| 1996 | 642631  |
| 1997 | 752028  |
| 1998 | 901294  |
| 1999 | 1056025 |
| 2000 | 1224092 |
| 2001 | 1420025 |
| 2002 | 1647976 |

- 2) Consider the following hypothetical data. Fit in a log linear regression model and interpret the estimated regression coefficient. Show all the calculations.

| Dependent Variable (Y) | Independent Variable (X) |
|------------------------|--------------------------|
| 189.8                  | 173.3                    |
| 172.1                  | 165.4                    |
| 159.1                  | 158.2                    |
| 135.6                  | 141.7                    |
| 132.0                  | 141.6                    |
| 141.8                  | 148.0                    |
| 153.9                  | 154.4                    |
| 171.5                  | 163.5                    |
| 183.0                  | 172.0                    |
| 173.2                  | 161.5                    |
| 188.5                  | 168.6                    |
| 205.5                  | 176.5                    |
| 236.0                  | 192.4                    |
| 257.8                  | 205.1                    |
| 277.5                  | 210.1                    |
| 291.1                  | 208.8                    |
| 284.5                  | 202.1                    |
| 274.0                  | 213.4                    |
| 279.9                  | 223.6                    |
| 297.6                  | 228.2                    |
| 297.7                  | 221.3                    |
| 328.9                  | 228.8                    |
| 351.4                  | 239.0                    |
| 360.4                  | 241.7                    |
| 378.9                  | 245.2                    |

*Adapted from Maddala (2001)*

- 3) In Example 9.4, the sample estimate of India's rate of growth during the period 1960-80 has been obtained to be about 3.4 per cent per year. Test the hypothesis that during the same period the rate of growth has in fact been 4 per cent per year.

## 9.13 KEY WORDS

### Assumptions of the Classical Regression Model:

- 1) The disturbance term  $U$  has a zero mean for all the values of  $X$ , i.e.,  $E(U) = 0$ .
- 2) Variance of  $U$  is constant for all the values of  $X$ , i.e.,  $V(X) = \sigma^2$ .

- 3) The disturbance terms for two different values of  $X$  are independent i.e.,  
 $\text{cov}(U_i, U_j) = 0$ , for  $i \neq j$ .
- 4)  $X$  is non-stochastic.
- 5) The model is linear.

|                                                    |                                                                                                                                                                                                                                                          |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Classical Normal Regression Model</b>           | : A linear regression model, in which, in addition to the five assumptions of the classical regression model, one more assumption of the error term being normally distributed is made.                                                                  |
| <b>Classical Regression Model</b>                  | : The conventional regression model whose parameters are estimated by the ordinary least square procedure.                                                                                                                                               |
| <b>Functional Forms of Linear Regression Model</b> | : Different kinds of linear regression models.                                                                                                                                                                                                           |
| <b>Gauss-Markov Theorem</b>                        | : Under the assumptions of the classical regression model, among all the linear unbiased estimators, the least square estimators have the minimum variance. In other words, the least square estimators are the best linear unbiased estimators or BLUE. |
| <b>Goodness of Fit</b>                             | : The ratio of the explained sum of squares to the total sum of squares. Also known as the coefficient of determination.                                                                                                                                 |
| <b>Hypothesis Testing</b>                          | : Conducting tests regarding hypotheses made about the unknown parameters of the population regression model with the help of the estimated sample regression function.                                                                                  |
| <b>Linear Regression Model</b>                     | : A regression model that is essentially linear in parameter.                                                                                                                                                                                            |
| <b>Non-Deterministic Regression Model</b>          | : A regression model in which the dependent variable is random or probabilistic but the independent variable is non-random.                                                                                                                              |
| <b>Parameters</b>                                  | : Unknown intercept and the slope coefficient of a population regression model.                                                                                                                                                                          |
| <b>Population Regression Function</b>              | : A linear regression model with a stochastic error term for a given population.                                                                                                                                                                         |
| <b>Sample Regression Function</b>                  | : The sample regression function that is fitted into the sample data for estimating the population regression model.                                                                                                                                     |
| <b>Standard Errors of Estimate</b>                 | : Standard deviations of the estimated sample regression intercept and slope coefficient.                                                                                                                                                                |
| <b>Two Variable Regression Model</b>               | : A regression model with one explanatory variable.                                                                                                                                                                                                      |

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## 9.14 SOME USEFUL BOOKS

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Gujrati, Damodar N. (2003); *Basic Econometrics*, McGraw-Hill, New York, U.S.A.

Maddala, G.S. (2002): *Introduction to Econometrics*, John Wiley & Sons (Asia), Singapore.

Pindyck, Robert S. & Rubinfeld, Daniel L. (1991); *Econometric Models & Economic Forecasts*, McGraw-Hill, New York, U.S.A.

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## 9.15 ANSWERS OR HINTS TO CHECK YOUR PROGRESS EXERCISES

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### Check Your Progress 1

- 1) See Section 9.2
- 2) See Section 9.3
- 3) See Section 9.4
- 4) See Section 9.6
- 5) See Section 9.7

### Check Your Progress 2

- 1) See Section 9.8
- 2) Measures the elasticity of the dependent variable with respect to the independent variable.
- 3) See Section 9.8
- 4) See Section 9.9
- 5) See Section 9.10

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## 9.16 ANSWERS OR HINTS TO EXERCISES

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- 1) 15.9 per cent
- 2) Do Yourself
- 3) Do yourself